



Prabhaskalyan — 23PH10028

Physics (BS 4Y)

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EDUCATION

Year	Degree/Exam	Institute	CGPA/Marks
2027	B.S, Physics (4YR)	Indian Institute of Technology, Kharagpur	8.4
2022	Intermediate Public Examination	Siddhartha Juinor College	91.6%
2020	Secondary School Certificate	Vijaya Bharathi English Medium School	100%

INTERNSHIPS

SDE Intern — Imago AI

Apr 2025 - Jun 2025

- Worked on Text-to-SQL models to convert natural language queries into SQL queries, improving the query generation process
- Developed agents to create dynamic dashboards and data visualizations using Python, streamlining the data using React.js
- Developed a Textual RAG system using **Pinecone**, integrated with a Text-to-SQL model to enhance accuracy in querying manufacturing SOPs
- Developing **LightSpeed** an automated ML model approval and testing system integrating AWS S3 to fetch test data and ML Model

AI Intern — MoneyClub

May 2025 - Aug 2025

- Implemented **Retrieval-Augmented Generation (RAG)** on a domain-specific knowledge base to improve accuracy and contextual relevance
- Developed end-to-end **WhatsApp automation** using AI for query handling, intelligent replies and solutions to users.

SDE Intern — Filo

May 2025 - Sep2025

- Developed a domain-specific language (DSL) to automatically generate AI-based educational tutorial videos from structured text prompts
- Integrated open-source tools like Manim, MoviePy, and OpenAI TTS models to synthesize narrated and visually animated content
- Streamlined the end-to-end video generation pipeline, enabling rapid prototyping and deployment of AI-driven educational content
- Developed an advanced AI-powered optics engine using Matplotlib to generate and visualize ray diagrams for complex optics problems

SDE Intern — PIN AI

Sep 2025 - Nov 2025

- Built a macOS-native Electron.js app integrating Gmail, Facebook, and MetaMask with secure on-device data storage with privacy-first archi
- Implemented a personalized AI system using RAG over local user data for context-aware responses using Ollma running LLMs locally
- Designed and deployed modular AI agents (Food, Travel, Shopping, Crypto) using a main-net / subnet architecture for scalable agents
- Developed trustless AI agents leveraging Ethereum ERC-8004 with on-chain validation and off-chain AI computation, deployed on AWS.

PROJECTS

Generative Engine Optimization — Distributed Task Processing

- Built distributed asynchronous task system using Celery and RabbitMQ for scalable workflow orchestration.
- Engineered multi-stage data pipelines coordinating generation, ranking, and persistence across worker nodes.
- Designed **fault-tolerant retry mechanisms** improving system resilience under high-throughput workloads.
- Implemented parallel processing with rate limiting using thread-lock synchronization to orchestrate LLM responses.

Registration Portal — Entrepreneurship Cell , IIT Kharagpur

- Exceeding **25,000 users** within 3 months, the portal successfully showcased its scalability and efficiency in handling the user influx
- Engaging more than **15,000 participants** from 20+ cities nationwide, the developed and implemented quiz proved highly successful
- Integrated payment gateway and REST APIs using Django and React,PSQL ensuring a smooth registration and transaction experience
- Developed an innovative referral system for Campus Ambassadors, leading to a remarkable and impactful **20% surge** in registrations

Hunger Quest — Self Project

- Developed the backend in Go and integrated PostgreSQL via GoOrm to manage data, group orders, and bookings with high performance.
- Implemented RabbitMQ for real-time shared cart updates, enabling low-latency synchronization of item additions and removals across all users.
- Utilized Kafka for event-driven communication in experience bookings and inventory updates, ensuring reliable and consistent data streams.
- Designed a microservices architecture with Docker (containerization) and Kubernetes (orchestration) to achieve scalability and LoadBalancing.
- Integrated Redis for caching to reduce latency and provide fast access to frequently used data such as images.

SKILLS AND EXPERTISE

Languages: C/C++, Python, JavaScript, Go,Node.js

Libraries and Frameworks: Numpy, Pandas,Matplotlib, Scikit-learn, Tensorflow, OpenCV, STL, Django, MERN stack,CNNs,RNNs,Transformers

Courses:Programming and Data Structures, Machine Learning, Deep Learning, Financial Modeling

POSITIONS OF RESPONSIBILITY

Associate Member— Entrepreneurship Cell — IIT Kharagpur

June 2024 - Nov 2024

- Collaborated on EAD Chennai 2024, driving event planning and execution, managing logistics, and coordinating with 23 Campus Ambassadors.
- Developed the EAD-LSM website using Django, overseeing design, implementation, and functionality, which has 25,000+ student visitors.
- Secured a partnership deal for a startup service program (SSP) , demonstrating strategic negotiation skills and business acumen
- Selected a team of 11 junior developers to work on the organization's projects, ensuring effective collaboration and successful delivery of tasks

Associate Manager— Entrepreneurship Cell — IIT Kharagpur

Dec 2024 - Aug 2025

- Ideated and successfully published an email newsletter that reached over 3,000 startups, leveraging Python and topmail for email automation
- Developed and deployed the GES 2025 website and a mobile app using React and React Native, ensuring a responsive, user-friendly design
- Managed a team of 11 junior developers, overseeing maintenance and development of portals and websites for smooth operations
- Led the successful execution of GES 2025, efficiently coordinating tech, operations, and outreach across teams during the 3-day summit
- Built a real-time judging portal for 12+ judges, saving 50% time with automated scoring and integrating Google Slides & OneDrive for pitch deck

Extra Curricular Activities

Actively contributed to National Service Scheme initiatives and the OpenClimateFix project, leading community activities and implementing SQL schema changes with new database fields.